

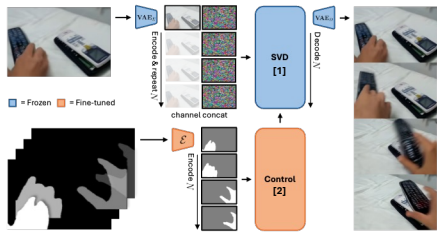
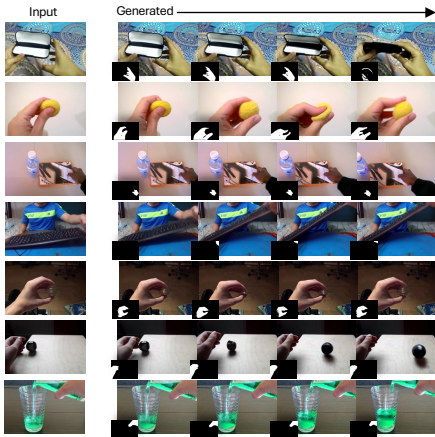
Overview

Can we generate physical interactions without explicit physics simulation?

Insight: Video generative models “distill” real-world video’s dynamics. We can exploit their “knowledge” of physics to “simulate” and render at the same time

Setup

Given an input image and a driving “action”, we generate a video of physical interaction within the scene

**Pipeline****Hand-Object Interactions**

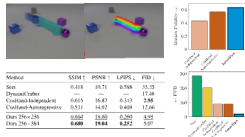
[3]

Frame and Motion Evaluation

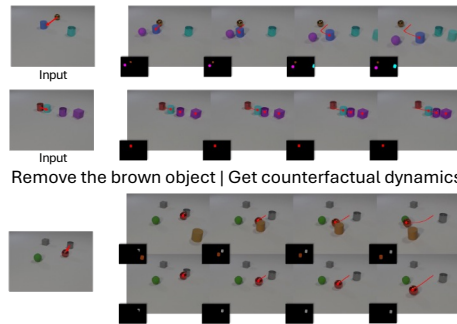
Measure the similarity between trajectories [4]

$$\frac{1}{m} \sum_{i \in \mathcal{T}} \max_{j \in \mathcal{T}} \text{corr}(\tau_i, \tau_j) + \frac{1}{n} \sum_{i \in \mathcal{T}} \max_{j \in \mathcal{T}} \text{corr}(\tau_i, \tau_j),$$

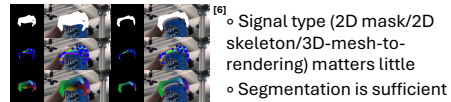
$$\text{corr}(\tau_i, \tau_j) = \frac{1}{\sqrt{v_i^2 + v_j^2}} \sum_{t=1}^T \frac{v_i^2 \cdot a_t^2 + v_j^2 \cdot b_t^2 + v_i^2 \cdot b_t^2 + v_j^2 \cdot a_t^2}{\sqrt{(v_i^2 + v_j^2)^2 + (v_i^2 + v_j^2)^2}}$$

**Physical Investigation**

Force propagation between uncontrolled objects



[5]

Ablation**Limitations**